

Cellular Research Track



Explore

Explore a real-world medical question,
developing the biological and experimental foundation needed to investigate it.

1

2

Customize

Customize your research pathway,
selecting the hands-on experiments you perform and curating your schedule.

☒	Cell Culture
☒	IF Stain
☐	Western Blot



Experiment

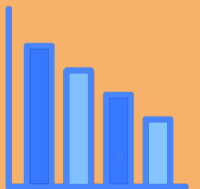
Perform hands-on laboratory experiments,
building real biotechnology skills using professional-grade equipment.

3

4

Interpret

Interpret your data,
transforming results into publication-quality scientific figures.



Publish

Synthesize your results,
presenting your findings in a collaborative research manuscript.

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Dive Deeper

Dive deeper each month,
repeating the research cycle with a new aim that strengthens the central conclusion.

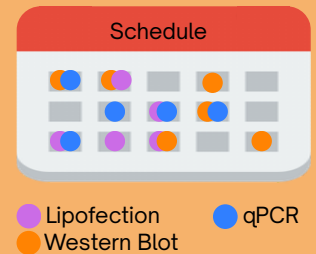


About the Program

Cellular Research Track is the first fully flexible research program that allows you to completely customize your learning path: choosing your own hours and the laboratory experimentation you wish to complete. Designed to cultivate researchers prepared for the biotech landscape, participants generate professional scientific data and publication-quality figures while investigating real medical challenges. These findings are ultimately synthesized into a collaborative research manuscript, earning participants verified technical badges and co-authorship recognition. Within the Cellular Research Track, each of the 4-months explores progressively deeper layers of the same central hypothesis, building a comprehensive understanding of how the biological pathway operates.

Flexible Research Model

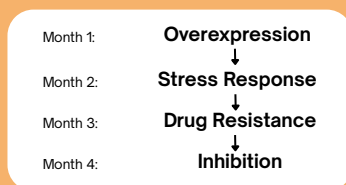
In-person laboratory sessions are fully modular: you customize your own track and book only the hands-on experiments you choose to perform. You only pay for the laboratory work you add to your portfolio, while remaining free to explore additional topics through the online course. Every participant receives full access to our comprehensive online curriculum, enabling independent mastery of the scientific concepts and protocols at your own pace.



Central Hypothesis:



Do stress-activated survival pathways enable cells to adapt and resist chemotherapy?



Hypothesis-Driven Investigation

Within a central hypothesis, participants address a focused research aim each month using advanced biotechnology techniques to generate measurable evidence, progressively uncovering the mechanistic connections within the pathway.

Collaborative Research Manuscript

Students learn to convert experimental data into publication-quality figures, developing skills in quantitative analysis and scientific communication. These figures are compiled into a collaborative research paper answering the central hypothesis. Authorship is designated according to meaningful contribution, including data collection, experimental work, and intellectual synthesis of results.

Eligibility

All students **MUST**:

- Be over the age of 18
- Complete our laboratory safety certification online course (\$30)
- Complete our in person laboratory orientation
- Submit signed user agreement and code of conduct

OpenLabs maintains a zero tolerance policy for sanitation and safety violations. Failure to follow these protocols results in immediate user review or expulsion.

Research Training Pathway

Each in-person laboratory module is structured to advance the central hypothesis through a distinct experimental layer, contributing evidence that builds toward the overarching collaborative research paper. Conceptual understanding also extends beyond the bench through the free online Research Track courses, which prepare students to perform protocols confidently, strengthen data interpretation skills, and provide tools for independent scientific exploration.

